

AI Resume Analyzer + Job Matcher

Complete Interview Concepts Checklist

This guide covers the concepts used directly or indirectly in the AI Recruitment Intelligence Platform: Resume Parsing, ATS Analysis, Semantic Matching, Embeddings, Gemini LLM Integration, Career Roadmaps, Mock Interviews, Recruiter Analytics, RAG Chatbot, FastAPI Backend, React Frontend, Databases, Docker, and Deployment.

A. Core Programming Foundations

- Python Syntax
- Functions
- Modules
- Packages
- Virtual Environments
- OOP (Classes, Objects, Inheritance, Polymorphism, Abstraction, Encapsulation)
- Exception Handling
- File Handling
- JSON Processing

B. Data Structures & Algorithms

- Arrays
- Strings
- Linked Lists
- Stacks
- Queues
- Hash Tables
- Trees
- Searching
- Sorting
- Time Complexity
- Space Complexity
- Big-O Analysis

C. DBMS & SQL

- Relational Databases
- SQL Queries
- Joins
- Indexes
- Normalization
- Primary Keys
- Foreign Keys
- Transactions
- ACID Properties
- Database Design

D. Backend Engineering (FastAPI)

- REST APIs
- HTTP Methods
- Routing
- Request/Response Lifecycle
- Pydantic Validation
- Middleware
- Dependency Injection
- API Documentation
- Async Programming
- CORS

E. NLP (Natural Language Processing)

- Text Cleaning
- Tokenization
- Stop Words
- Stemming
- Lemmatization
- Named Entity Recognition
- Information Extraction
- Skill Extraction

F. Resume Parsing

- PDF Parsing
- DOCX Parsing
- Text Extraction
- Structured Data Extraction
- Resume Section Identification

G. Artificial Intelligence & LLMs

- Generative AI
- Large Language Models
- Prompt Engineering
- Tokens
- Context Window
- Temperature
- Hallucinations
- Gemini API Integration

H. Embeddings & Semantic Search

- Vector Embeddings
- Dense Vectors
- Semantic Similarity
- Embedding Models
- Similarity Search
- Vector Representations

I. Resume Matching Engine

- Cosine Similarity
- Ranking Algorithms
- Recommendation Systems
- Semantic Job Matching
- Candidate Ranking

J. ATS Analyzer

- Applicant Tracking Systems
- Keyword Matching
- Resume Scoring
- Readability Analysis
- Action Verbs
- Quantifiable Metrics

K. Machine Learning Fundamentals

- Features
- Labels
- Training vs Inference
- Classification
- Regression
- Model Evaluation
- Overfitting
- Underfitting

L. RAG Chatbot

- Retrieval Augmented Generation
- Knowledge Retrieval
- Context Injection
- Grounding
- Vector Search Pipelines

M. Recruiter Analytics

- Leaderboards
- Skill Heatmaps
- Aggregation
- Filtering
- Dashboard Metrics

N. Frontend Development

- React
- JSX
- Components
- Props
- State Management
- Hooks (useState, useEffect)
- Routing
- API Calls
- TailwindCSS
- Responsive Design
- Framer Motion

O. Authentication & Security

- Authentication
- Authorization
- JWT Basics
- Password Hashing
- Environment Variables
- API Security

P. Docker & Deployment

- Docker
- Dockerfile
- Docker Compose
- Containers
- Images
- Port Mapping
- Render Deployment
- Vercel Deployment
- CI/CD Basics

Q. System Design

- Client-Server Architecture
- Frontend → Backend → AI Layer → Database Flow
- Scalability
- Caching Concepts
- Load Balancing Basics
- Microservices vs Monolith

R. PostgreSQL & SQLAlchemy

- ORM Concepts
- Models
- Relationships
- Migrations
- Connection Pooling

S. Must-Know Interview Questions

- Explain complete project architecture
- How resume parsing works
- How ATS scoring works
- What are embeddings?
- What is cosine similarity?
- How semantic matching works?
- How RAG works?
- Why FastAPI?
- Why PostgreSQL?
- Why Docker?
- How would you scale the application?

Priority Learning Order

- Python & OOP
- DSA
- DBMS & SQL
- FastAPI
- React Basics
- NLP
- Embeddings
- Cosine Similarity
- ATS Systems
- Machine Learning Basics
- RAG
- Docker
- System Design
- Deployment